

(varavi Ushakolla) 56.

Handwritten Reflection

1. Explain the difference between authentication and authorization?

- ⇒
- Authentication confirms the identity of a user (verifying who you are via username and password)
 - Authorization determines what users rights, permissions, and privileges that verified user has once inside the system (verifying what you can do)

3. What is the purpose of password hashing?

- ⇒ Hashing uses a one-way mathematical function to transform a plaintext password into a unique, fixed-length string of characters. It allows the server to verify credentials by comparing hash matching strings without ever storing the actual secret password itself.

2. Why should passwords never be stored in plain text?

- ⇒ Storing plain text passwords means anyone who gains unauthorized access to the database or filesystem can read every user's credentials.

immediately. This leads to severe identity compromise and credential stuffing attacks across other networks.

4. How does duplicate login prevention improve security?

⇒ It blocks active session hijacking and stops multiple unauthorized users from exploiting the exact same set of compromised user credentials at the same time.

5. Suggest two additional security improvements for your application?

⇒ (1) Wrap raw sockets using the python ssl library to implement Transport Layer Security (TLS/SSL) to encrypt all data travelling across the wire.
(2) Introduce cryptographic salting (adding unique random strings to passwords before hashing them) to completely break precomputed rainbow table database exploits.